Software Engineering — Project Guidelines

Fall 2026 — Scrum-Based Group Project

This document organizes the group project around three Scrum sprints (Sprint 0 for discovery and planning, Sprint 1 and Sprint 2 for build and iterate) plus a Final Demo.

1. Software Life Cycle and Tools

Life Cycle: Agile, Scrum, run across three sprints (Sprint 0, Sprint 1, Sprint 2) plus a Final Demo.

Project Type: Mobile or Web Application, custom development, UX-driven from wireframes.

Project Theme: Environment and Sustainability. Every team's product must be built around this theme — see Project Idea Guidelines in Section 4 for what's required beyond that.

- Backlog / Sprint Board: Trello
- Diagrams (ERD, Architecture): Draw.io or Visio
- Wireframes / UX: Figma, InVision, or Balsamiq
- Version Control: GitHub, GitLab, or Bitbucket (team picks one)

2. Project Expectations

- Two complete sprint implementations (Sprint 1 and Sprint 2), each with a working, demoable increment
- Complete wireframes for all screens, delivered before Sprint 1 build begins
- Complete ERD and Architecture Document, delivered in Sprint 0
- Backlog broken into epics and stories in Trello, updated sprint by sprint
- Evidence of real Scrum practice: sprint planning, daily scrum, and retrospective logs for every sprint

3. Milestones and Timeline

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Milestone	Est. Due Date	Focus
Team and Setup	Sept 17, 2026	Team formed (3 members, same section only), Environment-themed product idea + core features submitted for approval, Trello board and repo created
Sprint 0 — Discovery and Planning	Oct 2, 2026	Proposal, Requirements/SRS, Use Case Model and Descriptions, ERD, Architecture Doc, initial Trello backlog
Sprint 1 — Build and Review	Oct 31, 2026	Complete wireframes, first feature set implemented, Test Plan and initial Test Cases, Sprint 1 ceremony logs, Sprint 1 Review demo
Sprint 2 — Iterate and Complete	Nov 28, 2026	Remaining features implemented, test cases executed, Sprint 2 ceremony logs, Sprint 2 Review demo
Final Demo and Presentation	Nov 30 – Dec 12, 2026	Live working demo of full product, full backlog/sprint history walkthrough, Q&A with evaluators

4. Deliverables by Milestone

Team and Setup (due Sept 17, 2026)

Each team will conceive, design, and build a real mobile or web application of their own choosing over the course of the semester. The product should be a custom, UX-driven solution, not a clone of an existing app, that solves a genuine problem for a clearly defined set of users. Working through Sprint 0, Sprint 1, and Sprint 2, teams will take their idea from a written proposal through requirements, design, and two working increments, applying the Scrum practices covered in class along the way. The semester closes with a Final Demo, where each team presents a live, functioning product built end-to-end during the term.

- Team formed: 3 members, all from the same section (teams cannot include students from other sections)
- Project type selected: Mobile or Web Application — custom development, UX-driven from wireframes
- Product description: 3–4 lines explaining what the team will build, who it's for, and the problem it solves (see Project Idea Guidelines below)
- The core features the team plans to build, one line each — submitted for instructor approval before Sprint 0 begins
- Trello board and repository (GitHub, GitLab, or Bitbucket) created and shared

Project Idea Guidelines

Theme: every project must be built around Environment and Sustainability. The problem the app solves, and the users it serves, must be clearly tied to this theme.

Structural minimums: picking a theme doesn't by itself guarantee depth — a shallow, single-entity record-keeping app can be built on any theme. To make sure every idea has enough substance regardless of the specific concept chosen, every proposal must include:

- At least 3 distinct entities/resources, each with full Create-Read-Update-Delete (CRUD) screens
- At least 4 functions/features beyond account basics. Login, Logout, Signup/Register, Forgot Password, Change Password, and Profile all count as 1 feature combined — the remaining 3+ features must be substantial, theme-related functionality
- At least 1 multi-step transactional flow, where data carries across steps and the outcome depends on completing all of them — for example, a multi-step recycling pickup request (select items → choose a time slot → confirm), or a multi-step surplus-food handoff (list an item → match with a recipient → confirm pickup)

Along with the above, each team's Team and Setup submission must include a short written explanation covering:

- Who is the specific target user? (Not “everyone” or “people who care about the environment” in general — a clearly defined group.)
- What problem are they facing today, and how does the app solve it?

Sprint 0 — Discovery and Planning (due Oct 2, 2026)

- Project Proposal — interfaces (software/hardware/UX), features covered, cloud usage, server details, tech stack
- Requirements Specification (SRS) — functional and non-functional requirements / technical constraints
- Use Case Model and Use Case Descriptions
- Entity Relationship Diagram (ERD)
- Architecture Document — including git branching/workflow strategy
- Initial Product Backlog in Trello — epics and high-level stories

Sprint 1 — Build and Review (due ~Oct 31, 2026)

- Complete wireframes/UX for all screens
- Sprint 1 implementation — first batch of user stories built and working
- Trello: Sprint 1 backlog, stories, acceptance criteria, Definition of Done
- Test Plan and initial Test Cases
- Scrum ceremony logs: Sprint Planning, Daily Scrum, Sprint Review and Retrospective
- Sprint 1 Review — in-class demo of working increment

Sprint 2 — Iterate and Complete (due ~Nov 28, 2026)

- Remaining core features implemented (4 core features total across both sprints)
- Test Cases updated and executed
- Trello: Sprint 2 backlog groomed, stories, acceptance criteria, Definition of Done
- Scrum ceremony logs: Sprint Planning, Daily Scrum, Sprint Review and Retrospective
- Sprint 2 Review — in-class demo of working increment

Final Demo and Presentation (due ~Nov 30 – Dec 12, 2026)

- Live working demo of the complete product
- Full Trello backlog and sprint history walkthrough
- Code walkthrough and Q&A with evaluators
- Submission of all Scrum ceremony logs across both sprints

5. Grading Breakdown (Group Project = 30% of course grade)

Component	Points	What's assessed
Sprint 0 — Discovery and Planning	20	Proposal, SRS, ERD, Architecture Doc, initial backlog
Sprint 1 — Build and Review	20	Wireframes, first feature set, test plan, Sprint 1 demo
Sprint 2 — Iterate and Complete	20	Remaining features, test execution, Sprint 2 demo
Agile / Scrum Practice	20	Backlog health in Trello, sprint planning/daily scrum/retro logs across both sprints
Final Demo and Presentation	20	Working demo of full product, code walkthrough, Q&A